



Erosion has revealed the 65-million-year-old Cretaceous–Paleogene boundary—which corresponds to the extinction of the dinosaurs—in these rocks in the Hoodoo Formation, Alberta, Canada.

scientists concluded that neither people nor the environment had come to any harm.

Sony, a Japanese company, introduced a revolutionary new product in July: **the Walkman**. It was the first truly **personal portable audio device**, allowing people to carry their music with them on audio cassettes and listen to it on headphones.

Earlier in the year, the US company Bell Laboratories released the **seventh, and most important, version of UNIX**—a computer operating system that is the direct ancestor of the popular modern operating systems Mac OS X and Linux.

In October, British biologist **James Lovelock** published *Gaia: A New Look at Life on Earth*. The book set out **the Gaia hypothesis**. This states that Earth can be considered a single, self-regulating, living organism. Lovelock asserted that Earth's living things interact with the physical environment, keeping the oceans and atmosphere favorable for life to continue. Lovelock had been developing the idea and gathering relevant evidence since the early 1960s. The hypothesis influenced how many people tend to think about the environment, and the **interconnectedness and interdependence** of all living things in the world.



LUIS ALVAREZ (1911–88)

American physicist Luis Walter Alvarez's hypothesis about the cause of the extinction of dinosaurs came near the end of a long and illustrious career in physics. He made many contributions to radar technology during World War II, but his speciality was subatomic particles. In 1968, he won the Nobel Prize in Physics.

IN JANUARY, AMERICAN THEORETICAL PHYSICIST Alan Guth (b.1947) proposed a **refinement to the Big Bang theory** (see pp.344–45), which states that the Universe began

in an incredibly tiny, hot, dense state several billion years ago, and has been expanding ever since. He suggested that the **universe underwent "cosmic inflation,"** expanding by a factor of a trillion trillion trillion trillion (1 followed by 72 zeroes) in a tiny fraction of a second. Guth's proposition solved many existing problems with the Big Bang theory. Evidence in support of Guth's theory has since come from astronomers and particle physicists, and although some mysteries remain, it is now almost certain that cosmic inflation did occur.

In June, **Luis Alvarez** (see panel, left) put forward a theory to explain the **extinction of dinosaurs** 65 million years ago. Geologists had already noticed a distinct transition in rock layers dated to the time of extinction. Alvarez analyzed the rocks at this Cretaceous–Paleogene (K–Pg) boundary and found high levels of iridium, an element more common in asteroids. He suggested that **an asteroid**

Mushroom

The cholesterol-reducing compound lovastatin accounts for around 2 percent of the dry weight of edible oyster mushrooms (*Pleurotus ostreatus*), shown here.



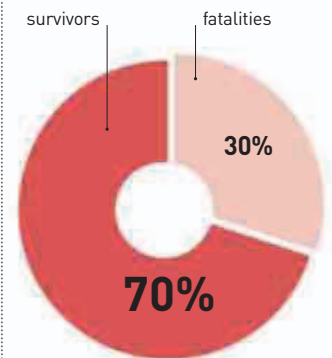
hit Earth, throwing up dust that blocked out the Sun for thousands of years, and leaving a still-visible crater. His hypothesis remained controversial until 1990, when a huge 65-million-year-old crater was discovered off the Yucatán peninsula, Mexico.

In the following month, pharmacologists announced the isolation of a compound called mevinolin from the mold (fungus) *Aspergillus terreus*. Mevinolin was shown to inhibit **the production of cholesterol**, which is associated with increased risk of heart disease. It was renamed lovastatin, and became the **first cholesterol-reducing "statin" drug** ever sold. In 1987, the US Food and Drug Administration (FDA) approved it, under the brand

name Mevacor. Lovastatin has since been discovered in other fungi, such as oyster mushrooms.

Earlier in the year, the thirty-third World Health Assembly declared that **smallpox had been eradicated globally**.

Space probe Voyager 1 made its **closest approach to Saturn** in November, flying within 77,000 miles (124,000 km) of the planet's cloud tops.



Deaths from smallpox

Until its global eradication in 1980, smallpox had claimed millions of human lives, killing about one-third of all those afflicted by it.

84 MINUTES
THE TIME IT TOOK **VOYAGER'S RADIO SIGNALS FROM SATURN TO REACH ASTRONOMERS ON EARTH**

October 4 British biologist James Lovelock publishes *Gaia: A New Look at Life on Earth*

December 24 The European Space Agency and ArianeSpace launch the first Ariane rocket

January 23 American theoretical physicist Alan Guth publishes his *Cosmic Inflation* theory

May 8 The World Health Assembly certifies the eradication of smallpox

June US company Seagate introduces the first 5.25-inch hard disk drive

June 6 American physicist Luis Alvarez publishes evidence to explain his theory on the extinction of dinosaurs

July Scientists announce they have isolated cholesterol-reducing lovastatin from the fungus *Aspergillus terreus*

November 12 Space probe Voyager 1 makes its closest approach to Saturn